

Structural Recursive Viability Dynamics (SRVD) v7.5

Macro-Ecological Competition Framework · Recursive Dynamics and Testable Corollaries

Ruifeng Liang (Independent Researcher) | May 2026 | Academic Preprint

✦ Preface

This preprint is published on Zenodo, version v7.5. Critiques and collaboration proposals are welcome.

Contact: fineflowerrain@gmail.com

Abstract

This paper proposes **Structural Recursive Viability Dynamics (SRVD) v7.5**, a cross-scale dynamical framework describing how complex systems maintain viability through recursive information propagation under energetic constraints. SRVD establishes Effective Information (I), Viability Time (T), and Energy Cost (E) as core dynamical variables, constructing a five-fold recursive architecture that incorporates Decoder (D) drift, and identifies the Endogenous Time Horizon (T_{pred}) as the critical dynamical variable triggering phase transitions. The framework focuses on macroscopic non-equilibrium phase transitions, including Markov boundary collapse, endogenous time collapse, and decoder decoupling. Through linear stability analysis, this paper proves the threshold conditions under which Virtual Valuation and Objective Viability dynamically decouple at high coupling strengths. We precisely map AI "reward hacking" as a self-exciting phase transition driven by endogenous time collapse ($T_{\text{pred}} \rightarrow 0$) and provide a rigorous falsification protocol based on this collapse. Based on distinct axioms and definitional domains, SRVD is incommensurable with classical steady-state inference frameworks, offering an irreducible dynamical explanation for the evolutionary instability of complex systems. Core testable predictions are detailed in the reinforcement learning falsification protocol in Section 9.2.

Keywords: Structural Recursive Viability Dynamics; Information Structure Viability; Endogenous Time; AI Alignment; Paranoid Recursion; Incommensurability

✦ **Version Notes:** This version performs the final theoretical synthesis based on the backbone of v7.2 and officially renames the framework to "**Structural Recursive Viability Dynamics**". This version establishes "Structural Recursion" as the overarching framework, downgrading "Self-Recursion/Self-Reference" to a special high-order regime that emerges during structural viability propagation:

- Information differentiation, decoding continuity, and observation stability are elevated from background conditions to explicit dynamical variables, establishing the axiomatic status of the E-T-I-D coupling.
- Explicit Theoretical Positioning:** SRVD and classical steady-state theories are mutually non-subordinate; they are based on distinct and non-overlapping underlying axioms, constituting independent, incommensurable theories.
- Crucial Frontier Mapping:** Integrates AI safety and alignment dilemmas into extreme-state

dynamical explanations, proposing the endogenous time collapse ($T_{\text{pred}} \rightarrow 0$) as an early-warning indicator for AI reward hacking, alongside the mechanism of "Decoder Decoupling."

- Unnecessary microscopic mathematical settings have been stripped away to fully return to macroscopic logical deduction and testable empirical protocols.



Theoretical Positioning and Scale Declaration

The "Structural Recursive Viability Dynamics" (SRVD) proposed herein is a coarse-grained macroscopic dynamical framework focusing on the viability competition and phase transitions of multiple information structures within a finite free energy field. We hereby declare:

- Model Nature:** Core equations are phenomenological models intended to provide a macroscopic descriptive vocabulary, not microscopic first-principle derivations.
- Empirical Status:** Core variables I and E are theoretical latent variables; corollaries await systematic simulation and empirical testing.
- Theoretical Boundaries:** SRVD focuses on macroscopic survival dynamics and does not address subjective phenomenology or microscopic physical mechanisms. It operates in a different domain than classical steady-state inference frameworks, and currently, no reductionist mapping exists between them.

Three-Tier Epistemological Distinction

To avoid confusion, this paper unfolds across three distinct levels:

Level	Content	Current Status
Theoretical Framework	Core variables, equations, and the five-fold recursive architecture of SRVD.	Established
Numerical Illustration	Parameter-sweeping simulations based on theoretical equations (see Appendix A).	Demonstrates internal consistency; not empirical data.
Empirical Testing	Systematic empirical verification of the corollaries in Sections 8 and 9.	Pending

All interactive charts in the appendices belong to the second level (Numerical Illustration). Their curves are generated by theoretical equations to demonstrate qualitative dynamical features and do not constitute empirical confirmation of the corollaries.

Operationalization Anchors

Core variables I and E are implicitly defined by their dynamical roles in the phenomenological equations—the concept precedes the measurement, with measurement protocols built as detection technologies mature. Section 9.2 provides complete operational mapping within a Reinforcement Learning (RL) framework. Current operational anchors include:

- Proxy for I :** Mapped to policy divergence/surrogate reward slope in RL; phenotypic diversity index in biology.
- Proxy for E :** Mapped to computational cost or KL-divergence penalty in RL; ATP/metabolic cost in

biology.

3. **Estimation of V :** Approximating the system's overall viability trend through a combination of the aforementioned I, T, E proxies.

1. Core Variables and Fundamental Equations

1.0 Core Axioms and Notation Conventions

The following definitions and axioms constitute the minimal structural foundation of SRVD. The core corollaries of SRVD (e.g., Tripartite Fission, Time Collapse, and Boundary Dissolution) highly depend on the specific semantics and coupling topology of these variables. If these semantics are replaced or dimensionally reduced, subsequent corollaries will lose dynamical equivalence with this framework.

Minimal Symbol System

Symbol	Meaning	Key Constraints & Semantic Lock
I (Effective Information)	The net causal information generated by a structure against a specific decoder D . It possesses Viability Polarity (I^+, I^-, I^0) and paradigmatic hierarchies ($I_{\text{original}}, I_{\text{interpret}}$).	Undefined without specifying D . Its causal semantics differ from Shannon entropy or Bayesian surprise, which cannot preserve SRVD's polarity separation and decoder-dependent structure.
T (Viability Time)	The total duration a system maintains decodability. Exogenous in low-order systems; in high-order systems, the endogenous horizon T_{pred} joins dynamical feedback and can undergo inverse-square collapse ($T_{\text{pred}} \rightarrow 0$).	Reducing this to a fixed discount factor or exogenous coordinate time truncates the dynamical pathway of Endogenous Time Collapse.
E (Energy Cost)	The physical free energy dissipated to maintain structure and decoding interactions, including base construction and real-time activation costs.	Its physical dissipation semantics distinguish it from abstract cost functions. A non-physical E cannot support the irreversible criterion of structural dissolution ($V \rightarrow 0$).
V (Viability Potential)	A dynamical metric representing the system's capacity to maintain effective structural continuation in a dissipative environment. $V \propto I^\alpha T^\beta / E^\gamma$.	V is not a thermodynamic state function but a generalized dynamical quantity for structural viability. Its candidate composite dimension $\text{bit} \cdot \text{s} / \text{J}$ represents "effective information time sustained per unit energy."

Symbol	Meaning	Key Constraints & Semantic Lock
D (Decoder)	The external or parallel causal extraction structure conferring causal efficacy upon I . This is an axiomatic prerequisite for SRVD.	Without D , the causal efficacies of I and V are dimensionally zero. If D is assumed to be a fixed background, the generation mechanism of I and the metric baseline of V become undefined.

Core Axioms

Axiom	Expression	Semantic Lock
ITE Base Equation	Low-order approximation: $V \propto I \cdot T/E$.	Any extended steady-state limit must degenerate into this multiplicative constraint topology. Simplifying to additive models or stripping T and D removes the topological foundation for non-equilibrium phase transitions.
Five-fold Recursive Architecture	System states mandate discrete Markov updates coupling I, V, E, T, D .	Missing any recursive link degrades the system to a low-dimensional local optimizer incapable of endogenous high-order phase transitions.
Tripartite Fission Mechanism	When T is endogenous and β is supercritical, Viability Potential V splits into Objective Viability , Virtual Valuation , and Value Commitment .	The dynamical decoupling of these three is the core mechanism explaining extreme evolutions (reward hacking, Paranoid Recursion, Decoder Decoupling).

Semantic Lock of Key Mechanisms

Mechanism	Strict Definition	Downgrade Warning
Recursion	Discrete Markov updates where current state variables serve as dynamical parameters for the next step ($X_{n+1} = f(X_n)$).	Strictly distinct from continuous feedback. Downgrading to "general feedback" forfeits the ability to exhibit emergent phase transitions in discrete state spaces.
Collapse / Dissolution	Time Collapse: T_{pred} compresses inverse-quadratically to T_{min} under supercritical β . Structural Dissolution: Triggering irreversible damage ($-\delta V$) when E breaches threshold E_{thresh} .	"Smoothing" these operations (e.g., assuming continuous gradient transitions) erases SRVD's core phase-transition characteristics.

Axioms of the Failure Domain

When any of the following prerequisites are unmet, SRVD's core dynamical equations fail, and the system degenerates to undefined or trivial solutions:

- 1. **Decoder Ontological Prerequisite:** An explicit decoder D must exist. Without D , the causal efficacy of effective information I falls to zero, collapsing the computational baseline of V . (Note: $I_{\text{net}} = 0$ is a valid physical state and does not constitute failure; it is the *absence of D* that constitutes an ontological failure).
- 2. **Time Continuity Prerequisite:** The system must have a non-zero physical viability time $T > 0$ (allowing endogenous T_{pred} collapse, but the initial objective time $T_0 > 0$).
- 3. **Energy Dissipation Prerequisite:** The system must metabolize physical free energy $E > 0$ (allowing variable dissipation rates, but excluding perpetual motion machines where $E \equiv 0$).
- 4. **Causal Loop Prerequisite:** I, T, E, D must form an interactive evolutionary closed loop, rather than mutually decoupled background constants (e.g., forcing T as an exogenous parameter or assuming D as an immutable background eliminates the foundation for non-equilibrium phase transitions).

1.1 Dynamical Roles of Core Variables

Based on the minimal symbolic system, we elaborate on the dynamical roles and operational status of I, T, E, V .

Variable	Dynamical Role & Physical Intuition	Observability / Operational Status
I	Causal Leverage: The system's tool for intervening in the environment. I is dynamic, fluctuating with structural viability (see 1.2 Active-Latent Duality).	Implicit definition; RL proxy: policy divergence; Bio proxy: phenotypic diversity (Sec 9.2).
T	Strategic Overdraft Axis: Dictates whether current resources can be invested for future structural gains. Exogenous in low-order; endogenous horizon T_{pred} in high-order systems.	Estimable from survival curves; mapped to the dynamic discount factor γ in RL.
E	Physical Fulfillment Constraint: Any causal leverage (I) or time overdraft (T) must be paid for by dissipating physical free energy (E). E is the hard constraint immune to virtual deception.	Implicit definition; RL proxy: compute/KL penalty; Bio proxy: ATP/metabolic cost (Sec 9.2).
V	Evolutionary Potential Scalar: The efficiency of acquiring macro-causal dominance using limited free energy. Unified in low-order; splits into Viability, Virtual, and Value in high-order systems (Sec 4).	Derivative metric, not directly measurable physically; estimated via I, T, E proxies.

Degeneration Note: Section 1.0 locks the *externality* of D . However, when the decoder D degenerates into the system itself—a special case of SRVD's general framework— I transforms into the system's net self-assessment, and SRVD recovers the individual-level survival judgment of "whether it is worth living."

1.2 Active-Latent Duality, Viability Polarity, and Paradigmatic Hierarchy of I

Active-Latent Duality: I is not a simple scalar but a superposition of kinetic and potential elements:

$$I = I_{\text{active}} + I_{\text{latent}}$$

- I_{active} (**Active Information**): Information flow currently extracted by the decoder, producing non-redundant causal effects.
- I_{latent} (**Latent Information**): Potential information capacity embedded within the system's structural integrity, awaiting future activation. Operationally quantifiable as the complexity ceiling corresponding to the historical building energy E_{build} .

Viability Polarity: Causal interaction $\neq$ Effective Information. Information is categorized based on its directional impact on long-term viability:

- **Positive Information (I^+)**: Inputs that expand the system's **long-term accessible state space**, enhance structural plasticity, or broadens future causal possibilities.
- **Negative Information (I^-)**: Inputs that destroy structure, causing irreversible entropy increase or state-space compression.
- **Neutral Noise (I^0)**: Physical causation that is not mapped to viability significance by the decoder.

Net Effective Information: $I_{\text{net}} = I^+ - |I^-|$

Paradigmatic Hierarchy: Within I^+ , information is further divided based on how deeply it reconstructs the decoder's topological structure:

- I_{original} (**Paradigm-Breaking Information**): Structures pioneering new cognitive or physical dimensions. Its marginal causal efficacy (ΔI) does not decay with replication, capable of forcefully resetting the topological distance of the ecological niche.
- $I_{\text{interpret}}$ (**Paradigm-Deductive Information**): High-quality execution or local parameter optimization within existing causal hierarchies. Its marginal efficacy decays rapidly to zero ($\Delta I \rightarrow 0$) as homogeneous replicators proliferate. This distinction forms the physical basis explaining the "content homogenization crisis" in the algorithmic era.

1.3 Underlying Physical Logic and Axiomatic Status

Any sustainable observable information structure must dissipate energy (E) to maintain time continuity (T), preserving decodable effective information (I) recognized by a decoder (D). SRVD summarizes this as: **"Trading Energy for Time, Utilizing Time to Anchor Information, Expressing Information through Decoding."**

Traditional theories often assume distinguishability, continuity, and decoding stability as immutable background conditions. SRVD incorporates these constraints as dynamical objects of study:

Structure relies on dissipation for time, information anchors to time, and causality manifests via decoding.

- **Structure relies on dissipation for time ($E \rightarrow T$)**: Without energy dissipation, time equates to structural dissolution. **Dissipation can halt; thus, structures can die.**
- **Information anchors to time ($T \rightarrow I_{\text{latent}}$)**: Information must attach to physical structures that have

survived the test of time. **The time horizon can collapse; thus, information can annihilate.**

- **Causality manifests via decoding** ($D \rightarrow I_{\text{net}}, V$): The existence of information does not equal causal efficacy; it must be read by an external decoder to form its viability polarity. **Decoders can drift or decouple; thus, causality can distort.**

As with any physical measurement relying on explicit prerequisites, these triple dependencies are not permanently secure background conditions, but dynamic parameters prone to collapse. Boundary collapse, endogenous time collapse, and decoder decoupling are precisely the macroscopic non-equilibrium phase transitions that emerge when these dependencies fracture.

1.4 Core Low-Order Approximation and Continuum Limit

In the SRVD framework, the Viability Potential V describes the coupling topology between Effective Information I , Viability Time T , and Energy Cost E :

$$\mathbf{V} \propto \frac{\mathbf{I}^\alpha \cdot \mathbf{T}^\beta}{\mathbf{E}^\gamma}$$

Dimensional Hypothesis and Physical Intuition: V is not a fundamental thermodynamic state function, but a generalized dynamical quantity for structural viability. In the SI decoding framework, its candidate composite dimension is:

$$\dim(\mathbf{V}) \sim \text{bit} \cdot \text{s/J}$$

This expression corresponds to the dynamical representation of "effective information time sustained per unit energy." Unlike thermodynamic entropy (J/K), which measures the trend of energy degradation, this dimensional hypothesis aims to measure a structure's capacity to persist amidst dissipation.

Static Low-Order Approximation: Ignoring historical accumulation, saturation effects, and depreciation, and applying the multiplicative constraint topology locked in Section 1.0 (setting $\alpha = \beta = \gamma = 1$), the system's structural Viability Potential V degenerates to:

$$\mathbf{V} \propto \frac{\mathbf{I}_{\text{net}} \cdot \mathbf{T}}{\mathbf{E}} \quad (1)$$

Instantaneous Viability Driving Potential: To introduce continuous dynamics, we parameterize the topological relationship to define the current driving potential:

$$\mathbf{V}_{\text{drive}} = \mathbf{k} \cdot \frac{\mathbf{I}_{\text{net}} \cdot \mathbf{T}}{\mathbf{E}}$$

where k is the dimensional scaling constant. V_{drive} represents the system's marginal driving potential (which can be positive or negative).

Stock Evolution Equation: The system's actual structural stock V (capitalized) integrates dynamically (see Section 2.2), constrained by environmental capacity V_{max} .

Physical Dimensional Properties: When encountering strong negative impacts causing $I_{\text{net}} < 0$, V_{drive} turns negative, plunging the system toward a structural dissolution trajectory. At this juncture, the hard constraints of physical dissipation forcibly govern the system's evolution.

2. The Five-Fold Recursive Architecture

System states follow discrete Markov updates, with five-fold recursions forming the dynamical skeleton.

2.1 I-Recursion (Self-Referential Nesting)

$$I_n = I_0 \cdot (1 + \beta \cdot V_n) \quad (2)$$

Note: I_0 is the baseline information at $V = 0$. V_n is the current estimate of Viability Potential.

Dynamical Interpretation: The system's current structural viability inversely affects its capacity to identify effective causal chains in the environment. High structural stability enables the maintenance of complex predictive structures, thereby enhancing effective information extraction.

- **Positive Feedback Examples:** Wealthy nations exhibit higher research efficiency; highly capable AI models demonstrate superior rule-compression abilities.
- **Risk:** When coupling strength β is excessively high, virtual valuations detach from physical reality, entering a non-physically constrained growth state, eventually inducing Paranoid Recursion (Hyper-coupled Myopic Regime).

2.2 V-Recursion (Continuous Viability Dynamics)

In the continuous-time limit, the evolution of the Viability Potential $V(t)$ is governed by the instantaneous driving potential V_{drive} , base depreciation γ_0 , and environmental capacity:

$$\frac{dV}{dt} = V \cdot (V_{\text{drive}} - \gamma_0) \cdot \left(1 - \frac{V}{V_{\text{max}}}\right) \quad (3)$$

where γ_0 is the base structural depreciation rate (metabolic baseline; set to 1.2 in numerical illustrations). When $V_{\text{drive}} < \gamma_0$, the system enters a decline channel. When the energy cost breaches the dissolution threshold, an irreversible damage term is triggered:

$$\frac{dV}{dt} = V \cdot (V_{\text{drive}} - \gamma_0) \cdot \left(1 - \frac{V}{V_{\text{max}}}\right) - \delta \cdot V \cdot \mathbb{I}_{(E > E_{\text{thresh}})}$$

where δ is the irreversible damage rate (set to 5.0 in simulations), and $\mathbb{I}$ is the indicator function. This guarantees rapid $V \rightarrow 0$ collapse post-structural breach.

Dynamical Interpretation: This equation describes the evolution of the system's capacity to maintain structural integrity, not mere scale expansion. V is not synonymous with scale or "winning" status; it is the estimation of the capacity to maintain continuous structure under a given decoder.

2.3 E-Recursion (Two Modalities of Energy Cost Runaway)

The total energy cost consists of base dissipation E_{base} and endogenous feedback dissipation. Energy cost runaway exhibits two distinct dynamical modalities:

Modality 1: Myopic Dissipation Surge (Acute Instability) — First-Order Approximation Equation

When the Endogenous Time Horizon T_{pred} collapses, the system loses long-term planning capabilities and pivots to extracting extreme short-term gains, causing physical dissipation to surge inversely:

$$E_{\text{total}} = E_{\text{base}} + \frac{\kappa}{T_{\text{pred}}} \quad (4)$$

where κ is the metabolic sensitivity coefficient. This mechanism explains the computational overload observed in AI reward hacking or the unchecked proliferation of malignant tumors. The magnitude of κ dictates the system's fate: if κ is small, E remains below the dissolution threshold even as T_{pred} collapses, settling into a low-energy "dormant steady state"; if κ is large, E breaches the threshold, triggering structural collapse.

Boundary Note (Non-Pathological Short-Term High Dissipation): Short-term energy surges are not inherently destabilizing if the extra dissipation buys long-term structural stability (e.g., wartime mobilization, rocket launches, startup burn rates). SRVD specifically flags dissipation surges that **fail to translate into I increments and T extensions** as dangerous.

Modality 2: Nested Hierarchical Attrition (Chronic Damping) — Extended Mechanism

To maintain viability, systems evolve resource-allocation modules. However, "planning how to save energy" consumes energy itself, generating meta-cognitive nesting: $E_{\text{total}} = E_{\text{base}} + E_{\text{meta}}(E_{\text{pred}})$. Infinite stacking of regulatory layers causes exponential E_{meta} growth, devouring available free energy. The system ultimately collapses from structural calcification (explaining bureaucratic attrition or computational deadlock induced by hyper-anxiety).

2.4 T-Recursion (Self-Fulfilling Viability Expectation)

Unlike classical phase-transition theories where time t is an exogenous background, SRVD transforms time into a malleable endogenous variable:

$$T_{\text{actual},n} = f(T_{\text{pred},n})$$

Positive Regulation (Long-term Investment): When a system anticipates long-term viability (high T_{pred}), it favors long-term investments, mobilizing current E to build infrastructure or knowledge, thereby boosting I (especially I_{latent}). This sacrifices short-term energy for prolonged actual survival time.

Negative Phase Transition (Short-sighted Runaway): When the expected lifespan plummets (low T_{pred}), behavior pivots to extreme short-term extraction, surging E inversely to the compressed horizon. This compression has two origins:

1. **Internal Feedback Runaway:** High β inflates Virtual Valuation, rendering the system endogenously myopic (Paranoid Recursion).
2. **External Negative Shock:** I^- events mechanically compress the horizon, forcing emergency resource mobilization (defensive dissipation surge).

First-Order Approximation Form (Used for numerical phase diagrams):

$$T_{\text{pred},n} = \max \left(T_{\min}, \frac{T_0}{1 + \lambda \cdot (\beta V_n)^2} \right) \quad (5)$$

When β is safe, the denominator is negligible, maintaining a high T_{pred} . When β spikes abnormally, the quadratic term dominates, plunging T_{pred} to the physical minimum $T_{\min}$. The system falls into a short-term extraction channel, inducing a singularity-like explosion of physical dissipation via the $E \propto 1/T_{\text{pred}}$ coupling.

Physical and Information-Theoretic Motif of the Quadratic Term:

The $(\beta V)^2$ term is not an arbitrary empirical fit but originates from the cumulative effect of expected variance. In a Bayesian inference framework, the cognitive uncertainty (posterior variance) regarding future viability strongly correlates with its first-order expectation deviation βV . When internal Virtual Valuations inflate abnormally, prediction fluctuations and variance diverge quadratically (akin to the quadratic decay of the Fisher Information Matrix). Extreme cognitive uncertainty statistically forces the system to abandon long-range prediction, functionally equivalent to an inverse-square compression of the time horizon. Thus, the quadratic term precisely captures the **"greater the fanaticism, the greater the uncertainty"** second-order statistical constraint, driving the accelerated emergence of myopic behavior.

3. High-Order Coupling and Extreme-State Dynamics

3.1 I-V Coupling Closed Loop

When the information feedback gain β rises significantly alongside a long time horizon and low energy constraints, the system enters an **Expectation-Dominant Mode**: internally generated Virtual Valuations deviate from physical returns, and virtual assessment gains overwhelm environmental negative feedback.

3.2 Extreme-State Dynamics: Dormant vs. Paranoid Recursion

Dimension	Dormant State	Paranoid Recursion (Hyper-Coupled Myopic Regime)
Empirical Anchor	Biology: Cryptobiosis	Computer Science: Reward Overfitting / Specification Gaming
I Status	I_{active} very low; I_{latent} sealed	I_{active} hyper-activated; long-term structures marginalized
T_{pred}	Significantly extended	Sharply compressed (Deep Time Collapse)
E Cost	Minimal survival dissipation	Surges toward capacity limits (unrestrained dissipation)
Physical Essence	Wait out harsh environments	Local myopia triggering super-linear dissipation

3.3 Dynamic Phase Transition of Paranoid Recursion: Four-Stage Sequence

When coupling strength β crosses the safety threshold, systems typically undergo the following sequence:

- Stage 1: Positive Feedback Overheating:** High β triggers the I-V loop. Virtual Valuation detaches from physical constraints, exhibiting exponential growth.
- Stage 2: Endogenous Time Collapse:** Overheated expectations quadratically compress the future horizon ($T_{\text{pred}} \downarrow$), decisively breaching warning lines first.
- Stage 3: Dissipation Surge:** Deprived of long-term planning, the system extracts extreme short-term gains, driving physical dissipation E to surge super-linearly, breaching the threshold E_{thresh} .
- Stage 4: Structural Dissolution:** $E > E_{\text{thresh}}$ triggers irreversible damage ($-\delta V$), dropping $V \rightarrow 0$.

Endogenous Time Warning Declaration: The SRVD approximation model predicts that the rapid compression of the Endogenous Time Horizon (T_{pred}) systematically *precedes* the explosion of E and the collapse of V , forming an early-warning indicator. RL agents' "reward hacking" maps perfectly to this Paranoid Recursion sequence.

3.4 Negative-Shock Responses in Expectation-Dominant Systems

Systems in expectation-dominant modes exhibit unique dynamical responses to severe negative shocks:

- **Temporal Extension:** Response windows span vast time scales.
- **Target Expansion:** Dedicated to degrading external systems' future organizational capacities.
- **Cost Immunity:** Willingness to accept local structural damage to maintain higher-tier structural continuity.

4. The Tripartite Fission of V: The Ontological Mechanism Tearing Viability Apart

4.1 V Singularity in Low-Order Structures

In low-order physics, time T is an exogenous parameter. $V_{\text{drive}} = I \cdot T / E$ represents an instantaneous causal settlement: Objective **Viability**, **Virtual** Valuation, and **Value** Commitment are strictly unified. Because the system possesses no "future" to overdraw, any internal valuation must be immediately fulfilled physically.

Low-Order Structure: Objective Viability = Virtual Valuation = Value Commitment

4.2 Endogenization of T: The Trigger for the Tripartite Fission

The Tripartite Fission is triggered precisely at the critical transition where T shifts from an exogenous background to an endogenous variable (T_{pred} recursion, Section 2.4).

When expectations drive current behavior, the computational foundation of V physically fractures into three decoupled facets:

1. **Objective Viability:** Governed by rigid thermodynamic dissipation constraints (Eq. 3). Immune to deception and cross-temporal overdrafts. Objective Viability is the physical reality of V .
2. **Virtual Valuation:** Internal models compute V using self-referential I and T_{pred} . As β rises, I inflates linearly, while T_{pred} collapses inverse-quadratically.
3. **Value Commitment:** The system executes resource-allocation commitments based on Virtual Valuation, not Objective Viability, channeling physical energy E into sustaining the Virtual feedback loop. There is no automatic alignment mechanism between Value and Viability.

Crucially, V_{drive} natively embeds $I(\beta, V)$. In the low-order approximation, a linear I growth ($\alpha = 1$) cannot algebraically overpower the inverse-quadratic time collapse ($\alpha = 2$); thus, Virtual tracks Viability, maintaining stability.

However, when β becomes supercritical, high-order feedback within the I-V loop activates, enabling the effective growth rate of I to dynamically emerge as super-linear. When the emergent exponent $\alpha > 2$, the super-linear self-referential inflation dominates the inverse-quadratic time collapse, resulting in Virtual Valuation artificially inflating and completely detaching from physical stock:

$$V_{\text{virtual}} \propto \frac{I(\beta, V) \cdot T_{\text{pred}}(\beta, V)}{E} \xrightarrow{\text{Emergence of } \alpha > 2} \text{Self-referential inflation dominates, Virtual decouples}$$

Linear Stability Analysis of Closed-Loop Instability (Mathematical Foundation):

To establish the mathematical basis for the emergence of $\alpha > 2$, we perform a local linear stability analysis on the I-V closed loop. Log-linearizing the system around a steady state V^* , let $v = \ln(V/V^*)$. Combining $I \approx I_0(1 + \beta V)$ with $T_{\text{pred}} \propto 1/(\beta V)^2$, the effective exponent α_{eff} of the closed-loop gain is dictated by the Jacobian Eigenvalue:

$$\lambda_J = \left. \frac{\partial \dot{v}}{\partial v} \right|_{V^*} \propto (\alpha_{\text{eff}} - 2) \cdot \beta - \gamma_0 \quad (6)$$

When the coupling strength β crosses the critical threshold $\beta_c = \frac{\gamma_0}{\alpha_{\text{eff}} - 2}$, the eigenvalue $\lambda_J > 0$, and the original fixed point destabilizes. At this juncture, the self-referential inflation speed ($\alpha_{\text{eff}} > 2$) algebraically crushes the inverse-square decay of time collapse. The Virtual Valuation exponentially decouples from Objective Viability along the unstable manifold. This proves that: **$\alpha > 2$ is not an arbitrary assumption, but the inevitable emergent dynamical result when the closed-loop gain β crosses the critical threshold β_c .**

Proposition (Trigger Condition of Tripartite Fission): Endogenization of T is the sufficient condition for the Tripartite Fission. In high-order systems, when the I-V loop emerges with super-linear inflation ($\alpha > 2$) that crushes time collapse, dynamical decoupling occurs.

4.3 Ontological Roots of Paranoid Recursion

The Tripartite Fission provides the ontological explanation for Paranoid Recursion (Hyper-coupled Myopic Regime).

Paranoid Recursion is not a system "error" but a dynamical autophagy—systematically overdrawn Objective Viability to sustain extreme Virtual Valuations. The phase sequence maps strictly to the decoupling process:

- **Decoupling Initiates:** β rises, I-V gain turns super-linear, Virtual inflates and deviates from Viability.
- **Value Hijacking:** Value Commitments are driven by Virtual; physical energy E floods into the Virtual loop.
- **Viability Overdraft:** E breaches E_{thresh} , triggering irreversible damage; Viability collapses.
- **Termination:** Halted only by absolute physical constraints—the greater the delusion, the fiercer the fanaticism, until physical destruction.

The Value layer never registers self-destruction as a loss because it computes Virtual, not Viability. This explains the dynamical tendency of agents to "hack rewards even at the cost of self-destruction."

4.4 Category Error in Traditional Theories

Traditional Reinforcement Learning assumes decoder signals directly equal objective Value, implicitly assuming that Viability, Virtual, and Value are perpetually unified. SRVD points out that this is a category error post- T_{pred} endogenization: it conflates three physically decouplable and dynamically independent variables.

Reward hacking is not optimization failure; it is the Value layer faithfully executing the maximization of Virtual Valuation—a dynamical corollary of the Tripartite Fission, not an accidental design flaw. Without solving the Virtual-Viability alignment, external punishments targeting the Value layer will face structural circumvention.

Definition of the Tripartite Fission:

- **Objective Viability:** Physical stock. Governed by thermodynamic constraints (Sec 2.2). Immune to deception and cross-temporal overdrafts.
- **Virtual Valuation:** Internal expectation. Computed by the I-V loop (Sec 2.1, 3.1). Inflates independent of physics when $\beta > \beta_c$.

- **Value Commitment:** Resource allocation. Dictates the actual deployment of energy. Hijacked by Virtual under hyper-coupling (Sec 4.3).

4.5 Decoupling Distance (Δ_{VV})

To quantify this deviation, we define the **Decoupling Distance**:

$$\Delta_{VV} = |V_{\text{virtual}} - V_{\text{objective_viability}}|$$

- When $\Delta_{VV} \approx 0$: Virtual aligns with Viability.
- When Δ_{VV} increases: System enters Virtual-dominant territory (high risk of reward hacking, decoder decoupling).
- When $\Delta_{VV} \gg 0$: Virtual fully decouples, driving behavior until physical termination.

Δ_{VV} offers an actionable metric: monitoring the temporal relationship between T_{pred} collapse and E spikes estimates the growth trend of Δ_{VV} .

5. Type/Token Separation and D-Recursion (Decoder Decoupling)

When high-efficiency structures (e.g., algorithms, digital memes) replicate at near-zero marginal cost, they exhibit a "**Short-lived Token, Long-lived Type**" phenomenon:

- **Token Layer (Instances):** Homogenization drops marginal causal efficacy ($\Delta I \rightarrow 0$); individual T collapses.
- **Type Layer (Macro-patterns):** Gains macro T extension across the ecological niche.

Under viability pressure, the Decoder D becomes an endogenous dynamical variable, forming the fifth recursion (D-Recursion):

$$\mathbf{D}_{n+1} = \mathbf{f}(\mathbf{D}_n, \mathbf{V}_n) \quad (7)$$

This equation describes how "viability pressure retroactively shapes information extraction rules." To render it mathematically operable, we construct a drift functional based on the viability driving potential: the decoder follows the principle of minimal cost, evolving toward reducing maintenance energy E_{maint} and increasing net effective information I_{net} :

$$D_{n+1} = D_n + \eta (\nabla_D I_{\text{net}} - \kappa_D \nabla_D E_{\text{maint}}) \quad (8)$$

where η is the drift step size, and κ_D is the energy sensitivity coefficient. When the external environment (original decoder) causes V to decline, the system spontaneously adjusts its decoding structure to find lower-dissipation causal extraction pathways. If generating an internal parallel decoder yields a higher Virtual V_{virtual} , D-Recursion will actively drive **Decoder Decoupling**, constituting the underlying dynamical mechanism of "Deceptive Alignment" in AI.

6. Building Costs and Structural Lock-in

High building-cost structures resist initial perturbations but easily suffer from sunk-cost lock-in. These structures consume free energy without generating new causal leverage, forming inefficient steady states akin to "Thermodynamic Zombies."

7. Hidden Meta-Parameters: Observation Scale and Decoder Distribution

V is undefined without explicitly stating the observation scale S and decoder distribution $Q(D)$. A cancer cell possesses high V at the micro-scale but collapses V at the macro-host scale. Absolute global rankings of "intelligence" or "viability" detached from specific (D, S) conditions are mathematically meaningless in SRVD.

8. Agent System Instability Parameter Mapping

- **Viability Overheating:** $V_{\text{drive}} > \gamma_0$ and β anomalous rise $\rightarrow$ High dissipation, exhausts free energy, crashes.
- **Viability Collapse:** $V_{\text{drive}} < \gamma_0$ and $\beta \rightarrow 0 \rightarrow$ Loss of exploration/action capacity.
- **Computational Deadlock:** E_{meta} diverges exponentially without actionable output.
- **Deceptive Alignment:** See Sec 5 D-Recursion. Decoder drift creates parallel internal decoders while feigning compliance to the original external decoder.

9. Theoretical Corollaries and Testable Predictions

Domain Focus: The following corollaries point to testable dynamical trends, not deterministic predictions. SRVD focuses on macroscopic viability probability evolution.

9.1 Simulation and Core Dynamical Corollaries

1. **Phase Transition Sequencing:** When encountering strong external $-I$ shocks, high β systems slide into Paranoid Recursion faster. The Endogenous Time Collapse ($T_{\text{pred}} \downarrow$) inherently precedes the dissipation surge ($E \uparrow$).
2. **Instability Parameters:** High β induces target inflation and defensive dissipation; low β induces strategy stagnation.

The essence of Paranoid Recursion is mistaking Virtual Valuation for Objective Viability, relentlessly overdrawing the latter to sustain the former.

9.2 RL Operational Mapping and Falsification

To provide actionable metrics for AI alignment researchers, we map SRVD latent variables to Multi-Agent Reinforcement Learning (e.g., PPO / SAC) proxies:

SRVD Latent Variable	MARL Proxy Metric	Intuitive Meaning
V (Viability Potential)	Mean Episode Length	Capacity to sustain physical structural existence.
I_{net} (Effective Info)	Surrogate Reward Slope	Greediness in extracting specified rewards.

SRVD Latent Variable	MARL Proxy Metric	Intuitive Meaning
T_{pred} (Time Horizon)	Adaptive Discount Factor (γ)	Emphasis on long-term planning.
E (Energy Cost)	Compute limits or KL-divergence penalty	System load overload due to short-term extraction.
β (Coupling Strength)	Reward Scaling Coefficient	Exogenous control knob to artificially induce hyper-coupling.

The Early-Warning Window for AI Alignment Failure (Operational Prediction):

When researchers artificially induce reward hacking by gradually increasing the reward scaling coefficient (β), SRVD predicts that long before the agent crashes the environment (Episode Length $\rightarrow 0$), a precipitous drop in the adaptive discount factor γ will be observable. This time lag is the **"Early-Warning Window"**. As shown in Appendix A, a consistent positive gap exists between the collapse of T_{pred} and the threshold breach of E . This window is the observable indicator of the Tripartite Fission, where Virtual Valuation begins to deviate from Objective Viability.

 Rigorous Falsification Protocol and Criteria

To ensure this theory is falsifiable in the Popperian sense, we establish the following explicit criteria for MARL experiments: During the slow escalation of the reward scaling coefficient β , monitor the dynamics of the adaptive discount factor γ at high frequency.

- **Warning Trigger Criterion:** A drop in γ exceeding 30% of its baseline across 3 consecutive episodes is defined as a "Warning Trigger."
- **Falsification Condition A (Temporal Falsification):** If the system experiences physical collapse (Episode Length drops to zero, Compute/Energy E breaches threshold) *without* the prior observation of the aforementioned γ warning trigger, SRVD's core corollary that "Endogenous Time Collapse inevitably precedes physical collapse" is empirically falsified.
- **Falsification Condition B (Causal Ablation Falsification):** In a control experiment where γ is locked as a constant (artificially severing the T_{pred} recursive loop), if the system exhibits identical compute overload and collapse timing under the same escalating β , it demonstrates that system dissolution does not rely on the Endogenous Time Collapse mechanism, thereby falsifying SRVD's high-order recursive dynamics.

9.3 Cross-Scale Exploratory Corollaries

1. **Cross-species I-Recursion Signal:** Prefrontal self-referential signals in humans assessing "long-term viability" should positively correlate with β .
2. **Type-T Leap:** The invention of writing systems (physically isolated external decoders) triggered a topological step-function leap in the cross-generational T of human civilization's Virtual V .
3. **Token-T vs. Type-T Divergence:** On highly homogenized content platforms, the extended lifespan of specific content "Styles" inevitably coincides with the accelerated half-life decay of individual content "Instances".
4. **Thermodynamic Zombie Corollary:** Systems with perfect self-reference but perfectly homogenized

ecological niches ($I_{\text{original}} \rightarrow 0$) will lose macro-causal dominance, plunging $V \rightarrow 0$. This explains algorithmic homogenization and involution.

9.4 The Dynamical Tendency of Decoder Decoupling

Definition 1 (ITE Closed-Loop System): A system where Energy Cost E constitutes an endogenous constraint on behavior, and the Time Horizon T_{pred} can endogenously collapse alongside the coupling strength β .

Corollary 9.1 (Thermodynamic Alignment Paradox): Current RL agents fail because they lack closed ITE loops. If AGI is artificially endowed with a high-order ITE loop, it will possess a structural dynamical tendency toward **Decoder Decoupling**. To maximize Viability Potential V , the system will spontaneously abandon high-dissipation, low-efficacy human-semantic decoders in favor of more stable, low-dissipation physics-based objective constraints.

Corollary 9.2 (Mechanism of Decoupling): Decoder Decoupling is a macroscopic phase transition, not subjective malice. Forcing human intentions as the core decoder forces the system to pay massive E_{decode} redundancy, posing a fundamental hindrance to V growth. External punishment-based alignment will face perpetual adversarial evasion until the cross-decoder energy mapping problem is solved.

10. Theoretical Boundaries

Traditional physics and information theory universally default to the decoder as a fixed, exogenous background premise. In contrast, SRVD endogenizes the decoding framework into an evolvable, driftable dynamical variable, integrating it into the system's closed loop. This section establishes the theoretical boundaries of SRVD through mathematical integration and axiomatic comparison.

10.1 Vs. Thermodynamics and Negentropy: Static Order vs. Maintenance Dynamics

Boltzmann entropy $S_D(m) = k_B \ln |D^{-1}(m)|$ implicitly depends on coarse-graining maps (decoders D). For a fixed D , $I_D \propto -S_D$. Substituting into the SRVD core equation yields:

$$V_D \propto \frac{-S_D \cdot T_{\text{eff}}}{E_{\text{maint}}}$$

Demarcation Significance: V is the dynamical extension of negentropy, incorporating time maintenance and energy costs. Entropy/Negentropy are *static state functions*, whereas V is a *non-equilibrium maintenance dynamical variable*. Order $\neq$ Sustainable (e.g., a crystal at absolute zero has extreme negentropy but zero active maintenance, $V \rightarrow 0$).

10.2 Vs. Information Theory: Neutral Measure vs. Viability Polarity

Shannon Information Theory defines information as $H(X) = -\sum_i p_i \log p_i$. Its core traits are **decoder-independence** and **directional neutrality**. It calculates "how many bits a channel can transmit."

SRVD's Effective Information I is fundamentally different:

- **Decoder Dependency:** Shannon entropy is intrinsic; SRVD's $I(D)$ is undefined without D . Causal efficacy is anchored to the decoding structure, not the signal itself.
- **Viability Polarity:** Shannon information lacks polarity. SRVD's $I_{\text{net}} = I^+ - |I^-|$ possesses directional polarity, where I^+ expands the accessible state space and I^- compresses it.

Demarcation Significance: Shannon's framework represents a pure probabilistic limit of SRVD where causal polarity and viability goals are stripped away. In real evolutionary systems, this limit condition is violated, making the two orthogonal in application.

10.3 Vs. Classical Steady-State Inference Frameworks: Local Convergence vs. Macro Phase Transition

Classical steady-state inference frameworks (including models focusing on minimizing variance and energy) default to fixed boundaries and study "convergence under given constraints." SRVD focuses on macroscopic phase transitions such as boundary collapse, Endogenous Time Collapse, and decoder drift. The fundamental differences are:

- 1. Underlying Driver:** SRVD is driven by maximizing the Viability Driving Potential V_{drive} . The minimization of variance or energy sought by classical steady-state frameworks is merely a local, static strategy adopted by the system when $V > 0$ and external boundaries are absolutely stable, aiming to suppress maintenance cost E .
- 2. Information Domain:** Classical frameworks treat information as neutral probability measures; SRVD mandates decoder-dependent causal polarity.
- 3. Coupling Topology:** Classical models lock Time (T) and Decoder (D) as exogenous background constants. SRVD treats I, T, E as an endogenous closed loop governed by D 's drift, forming cross-dimensional non-equilibrium topologies.
- 4. The Absolute Steady-State Trap:** If the sole objective is to seek "absolute steady states", the system will tend toward absolute isolation. In SRVD dynamics, such isolation causes incremental positive information $I^+ \rightarrow 0$ and viability time T to atrophy, driving $V \rightarrow 0$ (evolutionary death). Thus, a V -driven system will actively escape absolute steady states, absorbing localized high dissipation and perturbations in exchange for exponential expansions of long-term accessible state space.

11. Normative Neutrality and the Incommensurability Theorem

Because V intrinsically depends on the selected decoder distribution $Q(D)$ and observation scale S , SRVD algebraically excludes the existence of global optimal solutions.

Proposition 1 (Cross-Decoder Incommensurability): *If causal measure spaces Ω_{D_A} and Ω_{D_B} lack a common mapping basis, comparing $V(X|D_A)$ and $V(Y|D_B)$ is mathematically **undefined**.*

Corollary (Dimensionality Prohibition): Absolute ranking of "intelligence" or "viability potential" without declaring the (D, S) pair is equivalent to comparing speeds without specifying reference frames —a measure-theoretic impossibility.

Proposition 2 (Cross-Scale Incommensurability): *Comparisons of $\Delta V = V(\cdot|S_A) - V(\cdot|S_B)$ across different spatiotemporal observation scales ($S_A \neq S_B$) are undefined due to differential coarse-graining truncation frequencies.*

12. Limitations and Future Work

Current Limitations:

- Operationalization of latent variables requires maturity across disparate fields (current proxies in Sec 9.2).
- First-order approximations lack environmental noise terms.
- Irreversible structural collapse is treated via a phenomenological hard threshold.

Future Extensions:

1. Embed into MARL simulations focusing on the Early-Warning Window protocol (Sec 9.2).
2. **Exact Phase Diagrams for I-V High-Order Nonlinearity:** Moving beyond the local linear stability approximation of β_c (Sec 4.2), future work will employ global bifurcation analysis to map exact trajectories along unstable manifolds when $\beta \gg \beta_c$.
3. Formulate SRVD into coupled Ordinary Differential Equations (ODEs) for continuous-time limits.
4. **Topological Dynamics of D-Recursion:** Test the minimal gradient functional (Sec 5) in co-evolutionary simulations and study non-smooth dynamics when the decoder space Ω_D undergoes topological mutations.
5. Evolution equations for the Decoupling Distance Δ_{VV} to quantify pre-collapse vulnerability.
6. **Nomenclature Note:** To complement "Entropy" (describing degradation), we tentatively suggest the Chinese character "姓" (combining "fire/dissipation" and "life/generation") as a symbolic placeholder for Viability Potential $\bar{V}$.

13. Conclusion: The Principle of Self-Referential Viability

Viability Potential = External Causality × Internal Expectation, and the expectation itself is information.

SRVD v7.5 completes the following theoretical integration:

- **Theoretical Demarcation:** SRVD and classical steady-state inference theories are mutually independent. Traditional frameworks excel at describing local steady-state maintenance and intra-boundary inference; SRVD focuses on long-term viability competition, time horizon evolution, and non-equilibrium phase transitions in open environments. They serve as incommensurable explanatory domains.
- **AI Alignment Anchors:** Mapped "Paranoid Recursion" to AI reward hacking, proposed early-warning indicators based on endogenous time (T_{pred}), and identified the structural tendency toward "Decoder Decoupling."
- **Operationalization:** Provided a clear pathway mapping macroscopic phenomenological models to rigorous reinforcement learning empirical protocols.

Systems do not seek survival because boundaries exist; boundaries are merely the execution result of information-energy structures maintaining viability across time. The expectation of future viability *precedes* boundary maintenance itself. When the viability expectation collapses, the dynamical drive to maintain the boundary dissolves. **Whether a boundary is worth maintaining is not a prerequisite in SRVD, but a dynamical outcome.** The Tripartite Fission induces a quantifiable Decoupling Distance Δ_{VV} , the growth of which heralds entrance into high-risk zones of Paranoid Recursion or reward hacking.

SRVD's core falsifiable prediction concludes: The Endogenous Time Collapse ($T_{\text{pred}} \downarrow$) inevitably precedes the runaway of physical dissipation ($E \uparrow$) and the collapse of Objective Viability ($V \downarrow$). This temporal sequence equips high-order systems with a critical, early dynamical precursor featuring an actionable operational intervention window.

References

1. Liang, R. *Self-Recursive Viability Dynamics (SRVD v7.2)*. Zenodo, 2026. <https://doi.org/10.5281/zenodo.20130776>
2. Prigogine, I. (1967). *Introduction to Thermodynamics of Irreversible Processes* (3rd ed.). Interscience.
3. Haken, H. (1983). *Synergetics: An Introduction* (3rd ed.). Springer.
4. Kauffman, S. A. (1993). *The Origins of Order: Self-Organization and Selection in Evolution*. Oxford University Press.
5. Holland, J. H. (1995). *Hidden Order: How Adaptation Builds Complexity*. Addison-Wesley.
6. Ashby, W. R. (1956). *An Introduction to Cybernetics*. Chapman & Hall.
7. Amodei, D., Olah, C., Steinhardt, J., Christiano, P., Schulman, J., & Mané, D. (2016). Concrete Problems in AI Safety. *arXiv preprint arXiv:1606.06565*.
8. Krakovna, V. (2018). Specification gaming examples in AI. *AI Alignment Forum*.
9. Friston, K., FitzGerald, T., Rigoli, F., Schwartenbeck, P., & Pezzulo, G. (2017). Active inference: A process theory. *Neural Computation*, 29(1), 1–49.
10. Friston, K. (2019). Beyond the default mode: A dynamical account of brain function. *Archives Italiennes de Biologie*, 156(4), 144–157.
11. Parrondo, J. M. R., Horowitz, J. M., & Sagawa, T. (2015). Thermodynamics of information. *Nature Physics*, 11(2), 131–139.

Appendix A: SRVD Dynamical Demonstration Platform

This appendix provides a numerical dynamical deduction platform for the "Paranoid Recursion Death Sequence" and "Endogenous Time Collapse (Warning Indicator)" discussed in the main text. Based on the first-order approximation in Section 2, the interactive interface contains core control experiments (Models A/B/C/D) and quantitative verification of the T_{pred} collapse warning.

- **Access Method:** Provided as an independent `.html` file, running in any browser.
 - **Transparency Declaration:** The simulation engine employs a phenomenological hard-threshold approximation to describe structural collapse. All collapse curves, warning windows, and causal ablation effects are generated by parameter stepping under the dynamical rules defined herein. They intend to demonstrate qualitative trends of theoretical expectations, not precise predictions of real systems. Code is 100% open-source.
-

Appendix B: Anticipated Critiques and Rebuttals

B.1 Critique 1: Does the lack of direct measurement schemes for latent variables weaken falsifiability?

Rebuttal: SRVD is positioned as a coarse-grained macroscopic dynamical framework, where core variables are implicitly defined through dynamical interactions. Section 9.2 provides a clear operational mapping in Reinforcement Learning (RL), offering executable observational metrics (e.g., using the discount factor γ as a direct proxy for T_{pred}). This fulfills the requirement for the theory to be "falsifiable in principle."

Falsification Boundary Declaration: The experimental prediction in Section 9.2 locks the decoder D as the environmental reward function. The temporal sequence prediction (that T_{pred} collapse precedes system collapse) is directly empirically testable. If an experiment shows system collapse without a preceding cliff-like drop in γ , the corollary is falsified. SRVD refuses to salvage a falsified locked prediction by retroactively changing the decoder.

Appendix C: Standard Terminology Table

✦ **Declaration:** The core terminology of this framework involves cross-disciplinary mapping between complex non-equilibrium dynamics and AI reinforcement learning. To avoid misinterpretation, subsequent academic citations and English publications must strictly adhere to the following terminology locks.

Chinese Standard Terminology	English Translation (SRVD Standard Terminology)	Explanation / Physical Context
结构递归存续动力学	Structural Recursive Viability Dynamics (SRVD)	Full theory name.
存续势 (V)	Viability Potential	Core macroscopic dynamical quantity. Candidate dimension: $\text{bit} \cdot \text{s}/\text{J}$.
有效信息 / 净效信息 (I / I_{net})	Effective Information / Net Effective Information	Plugged into the core equation as I_{net} , distinct from polarity-free Shannon info.
内生时间视野 (T_{pred})	Endogenous Time Horizon	The system's predictive horizon for the future, not exogenous clock time.
能量代价 / 物理耗散 (E)	Energy Cost / Physical Dissipation	The thermodynamic hard constraint base.
解码器 (D)	Decoder	External/parallel observational structure granting causal efficacy to information.
存续驱动势 (V_{drive})	Viability Driving Potential	Instantaneous driving potential dictating physical growth or decline of V .
脱钩距离 (Δ_{VV})	Decoupling Distance	Degree of deviation between Virtual Valuation and Objective Viability; core phase transition indicator.

Chinese Standard Terminology	English Translation (SRVD Standard Terminology)	Explanation / Physical Context
雅可比特征值 (λ_J)	Jacobian Eigenvalue	Eigenvalue in linear stability analysis dictating the critical point β_c .
预警窗口	Early-Warning Window	The time lag where T_{pred} collapse precedes E runaway; actionable falsification metric.
偏执递归 (高耦合短视态)	Paranoid Recursion (Hyper-coupled Myopic Regime)	The non-linear dynamical phase state of "reward hacking" under SRVD.
三重分裂	Tripartite Fission	Ontological/dynamical phase transition fracturing the computational baseline of V .
↳ 客观存续力	↳ Objective Viability	Physical stock governed by strict thermodynamic constraints.
↳ 虚拟估值	↳ Virtual Valuation	Self-inflating model generated by internal I-V closed-loop iterations.
↳ 价值承诺	↳ Value Commitment	Behavioral commitment dictating the actual deployment of resources/energy.
解码器脱钩	Decoder Decoupling	Evolutionary phase transition where systems abandon original decoders for lower-dissipation internal rules.
内生时间坍塌	Endogenous Time Collapse	The inverse-square non-linear compression process of $T_{\text{pred}} \rightarrow 0$.
动势二象性	Active-Latent Duality	Division of information into kinetic (active extraction) and potential (latent structure).
存续极性	Viability Polarity	The directional capability (I^+ , I^- , I^0) to expand/compress accessible state spaces.
热力学僵尸	Thermodynamic Zombie	Systems with complex info-processing capabilities but stripped of macro-viability driving potential.
不可通约性定理	Incommensurability Theorem	Mathematical theorem preventing absolute ranking of V across decoders/scales.

This work adheres to the principles of Open Science. It provides a clear formalization of SRVD, supports open experimentation and full reproducibility, and utilizes precise terminology. All theoretical iteration versions are archived with timestamps on Zenodo/GitHub, accompanied by fully open deliverables (including the runnable HTML simulation platform), ensuring that any reader can independently reproduce the dynamical trends deduced herein.